Exercise 01 - SQL foundations

1. Retrieve all columns from the employees table

```
SELECT*
FROM employees;
```

ID	First-name	last-name	department	salary	hire-date	city
1	Alice	Green	IT	70000	2020-01-10	Johannesburg
2	Brain	Lee	HR	45000	2019-03-22	cape town
3	cathy	Zulu	Finance	65000	2018-07-18	Durban
4	David	Mokoena	Marketing	50000	2021-11-05	Pretoria
5	Eva	Naidoo	IT	72000	2017-09-30	Johannesburg

2. unique departments

```
SELECT DISTINCT department
FROM employees;
```

department
IT
HR
Finance
marketing

3. ~~Name~~ First and last name ordered by salary descending

```
SELECT first_name, last_name
FROM employees
ORDER BY salary DESC;
```

First_name	last_name
Eva	Naidoo
Alice	Green
Cathy	Zulu
David	Mokoena
Brain	Lee

4 Top 3 ~~highests~~ highest-paid employees

```
SELECT id, first_name, last_name, salary
FROM employees
ORDER BY salary DESC
LIMIT 3;
```

id	First-name	last-name	salary
5	Eva	Naidoo	72 000
1	Alice	Green	70 000
3	Cathy	Zulu	65 000

5. Employees in IT

```
SELECT id, first_name, last_name, department
FROM employees
WHERE department = 'IT';
```

id	first-name	last-name	department
1	Alice	Green	IT
5	Eva	Naidoo	IT

6 Finance with salary > 60 000

```
SELECT id, first-name, last-name, department, salary
FROM employees
WHERE department = 'Finance' AND salary > 60 000;
```

id	first-name	last-name	department	salary
3	Cathy	Zulu	Finance	65000

7 HR or Marketing

```
SELECT id, first-name, last-name, department
FROM employees
WHERE department = 'HR' OR department = 'Marketing';
```

id	First-name	last-name	department
2	Brain	Lee	HR
4	David	Mokoena	Marketing

8 Not in IT

```
SELECT id, first name, last-name, department
FROM employees
WHERE department = 'IT';
```

id	first-name	last-name	department
2	Brain	Lee	HR
3	Cathy	Zulu	Finance
4	David	Mokoena	Marketing

9 IT, HR, or finance using IN

```
SELECT id, first_name, last_name, department
FROM employees
WHERE department IN ('IT', 'HR', 'Finance');
```

id	First_name	last_name	department
1	Alice	Green	IT
2	Brain	Lee	HR
3	Cathy	Zulu	Finance
5	Eva	Naidoo	IT

10 IT with salary 265000 and city Johannesburg

```
SELECT id, first_name, last_name, department, salary, city
FROM employees
WHERE department = 'IT'
AND salary > 65000
AND city = 'Johannesburg';
```

id	First_name	last_name	department	salary	city
1	Alice	Green	IT	70000	Johannesburg
5	Eva	Naidoo	IT	72000	Johannesburg